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MODULE CODE: CET138

MODULE TITLE: Full Stack Development

MODULE LEADER: David Grey

SUBMISSION DATE AND TIME: Fri, 19 Jun 2026

ASSIGNMENT: 1

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- I understand that by submitting this assignment, I declare myself fit to be able to complete the assignment and I accept the outcome of the assessment as valid and appropriate.

CET138 Full Stack Development

HTML DOCUMENTATION

Details	
Name	Binod Katuwal
Student Id	250825048
Module Code	CET138
Module Title	Full Stack Development
Module Leader	David Grey
Submission Date	Fri, 19 Jun 2026
Assignment	Assignment 1 – portfolio (30%)
Project	Forever Moments – HTML Demo
Live Demo	https://binodkatuwal.github.io/Demos/htmldemo.html

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1. What is HTML?

HTML (Hypertext Markup Language) is the common structure language of the Web. It is NOT a programming language — it does NOT do logic or calculations, it describes the meaning of the content (headings, paragraphs, images, links, forms). HTML is the foundation for each website page. It is a structure that is filled with CSS (styling) and JavaScript (behavior). HTML5 has added the capabilities of semantic elements and native support for multimedia content.

2. Document Structure

Every valid HTML document follows a required structure. The <head> holds metadata (not visible to the user); the <body> holds everything the user sees.

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>My ePortfolio</title>
    <link rel="stylesheet" href="styles.css">
  </head>
  <body>
    <h1>Welcome</h1>
    <p>This is visible content.</p>
    <script src="app.js"></script>
  </body>
</html>
```

Element	Purpose
<!DOCTYPE html>	Declares HTML5 — must be the first line
<meta viewport>	Essential for responsive design on mobile devices
<title>	Text shown in the browser tab; used by search engines
<link rel="stylesheet">	Links an external CSS file
<script>	Links JS; placed at end of body for performance

3. Semantic HTML5

Semantic elements describe the meaning of content, not just its appearance — improving accessibility (screen readers), SEO, and code readability.

```
<header><nav>...</nav></header>
<main>
  <article>
    <h1>Page Title</h1>
    <section><h2>About</h2><p>...</p></section>
  </article>
  <aside><h2>Related Links</h2></aside>
</main>
<footer><p>&copy; 2024 My Portfolio</p></footer>
```

Element	Meaning
<header>	Intro content — logo, title, navigation
<nav>	Block of navigation links
<main>	Primary unique content — one per page
<article>	Self-contained content (blog post, card)
<section>	Thematic grouping with its own heading
<aside>	Sidebar / related content
<footer>	Copyright, contact, links

4. Links, Images, Lists and Tables

```

<!-- Links -->
<a href="https://example.com" target="_blank" rel="noopener">External</a>
<a href="about.html">Internal Page</a>
<a href="#contact">Jump to Section</a>

<!-- Image with required alt text -->


<!-- Unordered & ordered lists -->
<ul><li>HTML</li><li>CSS</li><li>JavaScript</li></ul>
<ol><li>Plan</li><li>Build</li><li>Style</li><li>Add JS</li></ol>

<!-- Table -->
<table>
  <thead><tr><th scope="col">Item</th><th scope="col">Weight</th></tr></thead>
  <tbody><tr><td>Assignment 1</td><td>30%</td></tr></tbody>
</table>

```

5. Forms

Forms allow users to input and submit data. Correct input types and label associations are essential for usability and accessibility.

```
<form action="/submit" method="POST">
  <label for="name">Name:</label>
  <input type="text" id="name" name="name" required minlength="2">

  <label for="email">Email:</label>
  <input type="email" id="email" name="email" required>

  <label for="msg">Message:</label>
  <textarea id="msg" name="msg" rows="4"></textarea>

  <fieldset>
    <legend>Contact Method:</legend>
    <label><input type="radio" name="c" value="email"> Email</label>
    <label><input type="radio" name="c" value="phone"> Phone</label>
  </fieldset>

  <button type="submit">Submit</button>
</form>
```

Input Type	Use
text / email	Names, usernames / validated email format
password	Masks characters for secure entry
radio / checkbox	Single choice from group / multiple selections
date / number	Native date picker / numeric with min, max
submit / reset	Sends form data / clears all fields

6. Accessibility & Best Practices

- Always provide alt text on images for screen readers and SEO
- Use semantic elements (header, nav, main, footer) instead of generic `<div>`
- Associate every form input with a `<label>` using `for/id`
- Use one `<h1>` per page and a logical heading hierarchy (`h1 → h2 → h3`)
- Never use tables for layout — only for tabular data
- Validate HTML with the W3C Validator to catch errors

7. Summary

HTML is the structural backbone of the web — every page begins with it. This section demonstrates a thorough understanding of the core building blocks of HTML:

- The required HTML5 document structure and metadata
- Semantic HTML5 elements for accessible, well-structured pages
- Links, images, lists, and tables with correct semantic markup
- Forms — input types, labels, field sets, and validation attributes
- Accessibility and best-practice principles for clean, maintainable HTML